

Eclipse IDE :

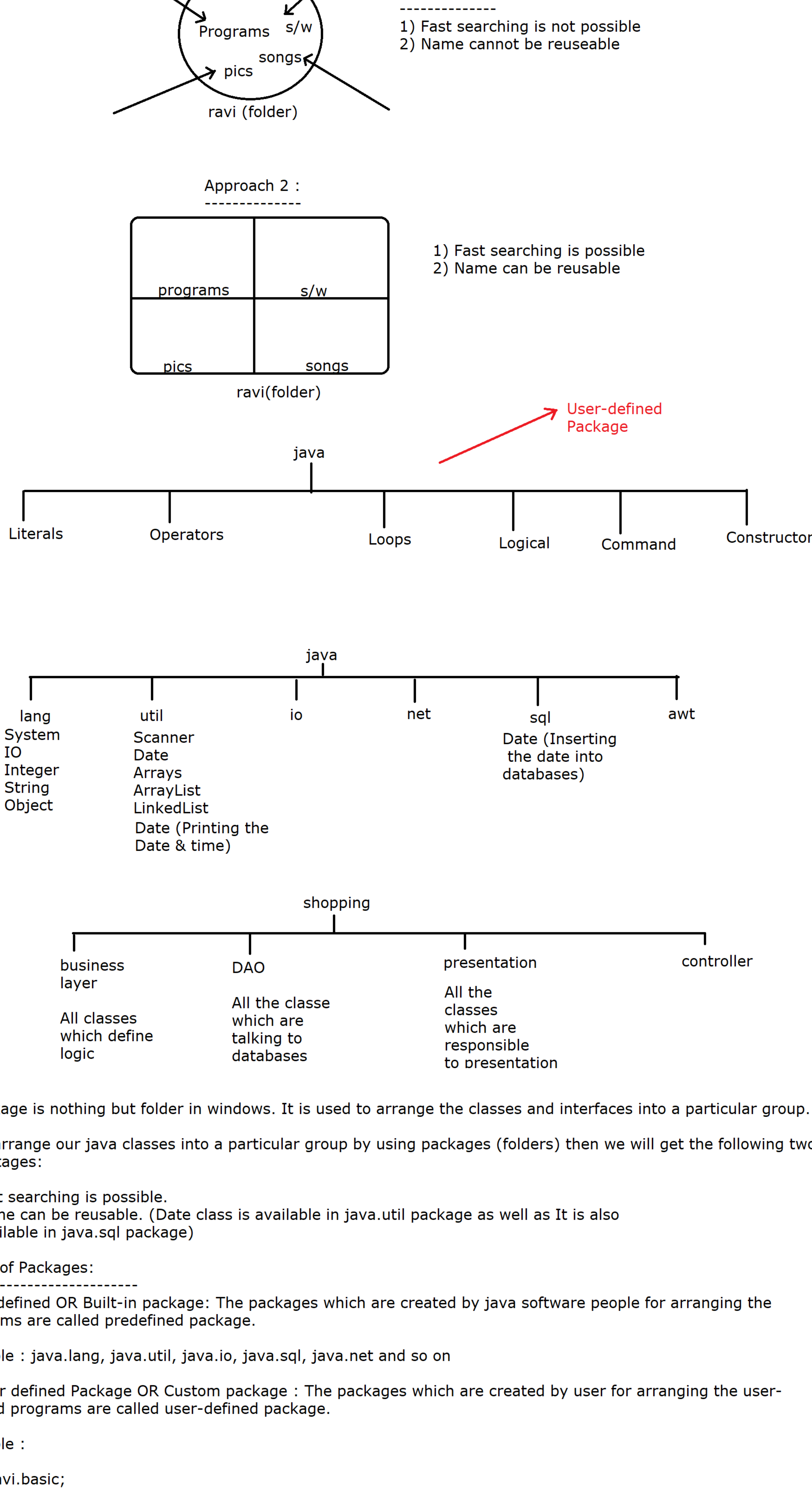
IDE stands for "**Integrated Development Environment**". By using eclipse IDE, we can develop, compile and execute our java program in a single window.

The main purpose of Eclipse IDE to **reduce the development time**, once the development time will be reduced then automatically the **cost of the project** will be reduced.

How to create a Project in Eclipse IDE:

File -> new -> Project OR Java Project -> Provide the name for the project (Batch 60)-> Finish

What is a Package in Java ?



A package is nothing but folder in windows. It is used to arrange the classes and interfaces into a particular group.

If we arrange our java classes into a particular group by using packages (folders) then we will get the following two advantages:

- 1) Fast searching is possible.
- 2) Name can be reusable. (Date class is available in java.util package as well as It is also Available in java.sql package)

Types of Packages:

- 1) Predefined OR Built-in package: The packages which are created by java software people for arranging the programs are called predefined package.

Example : java.lang, java.util, java.io, java.sql, java.net and so on

- 2) User defined Package OR Custom package : The packages which are created by user for arranging the user-defined programs are called user-defined package.

Example :

```
basic;  
com.ravi.basic;  
com.tcs.online_shopping;
```

What is BLC & ELC class ?

BLC class :

- * It stands for "Business Logic class".
- * Here We will write logic for our java program.
- * BLC class will never contain main method. [It will only contain Logic Part]

Example :

```
//BLC  
public class Calculator  
{  
    public static void sum(int x, int y)  
    {  
        IO.println("Sum is :"+(x+y));  
    }  
  
    public static void sub(int x, int y)  
    {  
        IO.println("Sub is :"+(x-y));  
    }  
}
```

ELC class :

- * It stands for "Executable Logic Class" OR Driver class.
- * It must contain main method.
- * The execution of the program will start from main method only that means from this class only.

Example :

```
public class Main  
{  
    void main()  
    {  
        //Execute the program from Here.  
    }  
}
```

How many public classes we can write in a single java source file?

We can write only one public class in a java source file and It must be our file name.

Case 1:

```
Welcome.java    //Here Welcome class must be a public class
```

```
public class Welcome  
{  
}
```

Case 2:

```
Test.java  
public class Test  
{  
}
```

```
public class Welcome //CE Here Welcome class must not be public  
{  
}
```

Case 3:

```
Test.java  
public class Test  
{  
}
```

```
class Welcome //It is valid  
{  
}
```

```
Circle.java  
public class Circle //public  
{  
}
```

```
class Square //non public  
{  
}
```

folder
(com.ravi.blc)

```
Main.java  
public class Main  
{  
}
```

folder
com.ravi.elc